# Radiation Criteria

- In Tile Calorimeter, the fingers, where LVPS bricks are located at, are the harshest location in terms of radiation
  - The very early design of the bricks were showing excessive trips, so a customized rad-hard design for these bricks is essential
- Based on the new guidelines and simulations performed in 2020, the radiation criteria were updated

| Radiation Type | Location | Simulated Dose/ Fluence | $SF_{sim}$ | $SF_{other}$ | Target Dose/ Fluence |
|---|---|---|---|---|---|
| Total Ionizing Dose (TID) [Gy] | Barrel | 53.6 | 1.5 | 5 (Low dose rate effects) | 402 (80.4 for CMOS) |
| | End-caps | 34.3 | 1.5 | 5 (Low dose rate effects) | 257.2 (51.5 for CMOS) |
| | Average | 43.9 | 1.5 | 5 (Low dose rate effects) | 329.3 (65.9 for CMOS) |
| Non-Ionizing Energy Loss (NIEL) [$n/cm^2$] | Barrel | $3.66 \times 10^{12}$ | 1.5 | 1.3 (Mono-energetic beam) | $7.14 \times 10^{12}$ |
| | End-caps | $9.58 \times 10^{11}$ | 1.5 | 1.3 (Mono-energetic beam) | $1.87 \times 10^{12}$ |
| | Average | $2.31 \times 10^{12}$ | 1.5 | 1.3 (Mono-energetic beam) | $4.5 \times 10^{12}$ |
| Single Event Effects (SEE) [$p/cm^2$] | Barrel | $5.3 \times 10^{11}$ | 1.5 | 2 ($E_{cut}$ = 20 MeV) | $1.59 \times 10^{12}$ |
| | End-caps | $9.8 \times 10^{10}$ | 1.5 | 2 ($E_{cut}$ = 20 MeV) | $2.94 \times 10^{11}$ |

- As the end-cap numbers are lower, barrel numbers were our criteria
  - If the bricks are swapped between barrel and end-caps in the middle of HL-LHC, the average of the two numbers can be considered as the criteria